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Deep learning-based identification of facial asymmetry for medical diagnosis

זיהוי חוסר סימטריה בפנים כתוצאה מבעיות רפואיות

פרויקט מחקרי

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הוכן לשם השלמת הדרישות לקבלת

תואר ראשון במדעי הטבע B. Sc.

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הוגש למחלקה למדעי המחשב

המכללה האקדמית להנדסה סמי שמעון

באר-שבע

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Table of Content:

Abstract _____ 1

1. Introduction _____ 2
2. Background _____ 3
3. Related work _____ 5
4. Method _____ 6
5. Data _____ 8
6. Experimental setup _____ 9
7. Results and discussion _____ 10
8. References _____ 11

Abstract:

Facial asymmetry identification is crucial for diagnosing and rehabilitating medical conditions such as facial paralysis, including Bell's palsy. Recent advancements in deep learning, particularly convolutional neural networks (CNNs), enable highly accurate and efficient object detection methods. This project explores CNN-based models for detecting facial asymmetry, comparing different network architectures to determine the most effective approach. By applying multiple CNN structures to the same dataset, we evaluate their performance and select the optimal model for accurate and reliable asymmetry detection. In addition, this project compares different face parts to choose the part that identifies facial asymmetry better. Our findings contribute to the development of automated, deep learning-based diagnostic tools that can enhance medical assessment and treatment planning.

Keywords: facial asymmetry detection, deep learning, convolutional neural networks, medical image analysis

1. Introduction:

The classification of facial asymmetry is an important yet challenging problem in health care informatics owing to its diversity, varying symptoms and complex underlying mechanisms. Adequate feature learning plays a vital role to enhance the accuracy of classification systems. The previous methods to encode facial palsy in terms of asymmetric facial features can be broadly classified as hand-crafted features based or learning-based methods. The hand-crafted methods rely on prior knowledge to extract the underlying asymmetrical features.

Levels of facial symmetry have been shown to indicate important information in areas like perceived attractiveness [4] and also in medical diagnosis and rehabilitation for conditions such as facial palsy or stroke. Specifically, within the medical domain the importance of diagnosis and rehabilitation tracking for the best possible outcomes in patients with facial paralysis such as bell's palsy has been highlighted within recent research [5–6].

Facial palsy is a common neuromuscular disorder causing facial weakness and the disability of facial expressions [1]. Palsy patients lose control of the affected side of their face, experiencing the dropping or stiffness of muscles and disorders of taste buds. Statistics regarding facial palsy report 25 incidents annually per 100,000 people, or approximately one patient out of 60 people in their lifetime, while an average of 40,000 palsy patients are reported in the United States every year [2]. Even though palsy does not cause patients to be in physical pain, they experience phycological stress, external discomfort, and depression, since palsy affects their appearance, facial movements, feeding functions, and, thus, their daily lives [3].

Traditionally, the diagnosis of facial palsy is clinically performed by specialized neurologists who force patients to perform specific facial expressions for evaluating the condition of certain face muscles. The level of palsy is assessed by evaluating the symmetry between the right and left parts of the face in terms of various scoring standards and measuring distances between facial landmarks for both sides with a simple ruler. Recent advancements in image analysis algorithms, combined with the increasingly affordable cost of high-resolution capturing devices, resulted in the development of efficient, simple, and cost-effective vision-based techniques for medical applications, reporting impressive state-of-the-art performances. The diagnosis of various diseases is greatly assisted by facial abnormalities recognition using computer vision, dynamically incorporating facial recognition into artificial intelligence (AI)-based medicine. Automatic image-based facial palsy could accelerate the diagnosis and progress evaluation of the disease, offering a non-invasive, simple, and time- and cost-saving method that could be used by the palsy patients themselves without the presence of a human expert.

Most image-based techniques are based on hand-crafted features associated with the affected facial regions, used as training data for machine learning and deep learning models. Such facial landmarks need to be accurate and discriminative for improving the performance of the models.

You will see in this project a fair comparison between different deep learning techniques. To make things much clearer the main deep learning model that the methods use is CNN.

2. Background:

Symmetry of the face is one of the most important features for perception of attractiveness. Facial asymmetry is common in humans. Significant facial Asymmetrical as well as esthetic problems cause both functional as well. When patients complain of facial asymmetry, the underlying cause should be investigated. Etiology includes congenital disorders, acquired diseases, and traumatic and developmental deformities.[7]

Facial nerve parlays: We focus on the most common neurological causes of facial asymmetry. Facial asymmetry can result from disturbances of the cranial nerves, developmental disorders, or myopathies. In the broad range of differential diagnostics of facial asymmetry, neurological abnormalities should be taken into consideration.[8]

Face recognition: face recognition recently received significant attention, especially during the past several years [9]. Face recognition is important research problems for various applications like security, authentication, healthcare. CNN is the most used model in deep learning.

Neural network: Creating a way to recognize patterns is extremely difficult to automate but appears to be performed very easily by humans. For instance, humans recognize various objects and make sense out of the large amount of visual information in their surroundings, apparently requiring very little effort. After researching they found Neural networks. The neural network of a human is part of its nervous system, containing many interconnected neurons (nerve cells). "Neural" is an adjective for neuron, and "Network" denotes a graph like structure.[12]
Artificial Neural Networks: refers to computing systems who's central The theme is borrowed from the analogy of biological neural networks.[6]

Convolutional Neural Network (CNN):
A Convolutional Neural Network (CNN) is a deep learning model specifically designed for processing images or videos. Unlike traditional neural networks, CNNs are particularly adept at detecting patterns like edges, textures, and shapes in images, making them ideal for tasks such as image recognition, classification, and segmentation. The architecture of CNN consists of four layers: convolutional layers which perform convolution, rectified linear unit (ReLU) layers which apply element wise activation function, pooling layers which perform down sampling operation, and fully connected layers (FC) which compute scores of a class.[10]
Pooling layer [13]: In the classical architectures of convolutional neural networks, convolution layers are followed by sub sampling layers. Sub-sampling layer reduces the size of cards and introduces invariance to (low) rotations and translations can appear as input. The output of max-pooling layer is given by the maximum activation value, in the input layer for different regions of size $Kx \times Ky$ non-overlapping. Similarly to a convolution layer, a bias is added, and the result is passed to the transfer function ϕ defined above.
Fully connected layer [13]: After several max pooling and convolutional layers, the high-level reasoning in the neural network is done fully connected layers. Neurons in a fully connected layer have full connections to all activations in the previous layer, as seen in regular Neural Networks.

Confusion Matrix: A confusion matrix is a table that summarizes the performance of a classification model by showing the counts of actual vs. predicted labels.

	Predicted positive	Predicted negative
Actual positive	True Positive (TP)	False Negative (FN)
Actual negative	False Positive (FP)	True Negative (TN)

Class: a class refers to a category or label assigned to data in a classification problem. It can be a binary like true or false or multi classification problem like facial asymmetry degree.[7]
Prediction: The term prediction is the class that you received for some input from the machine.
True Positive (TP): Correctly predicted positive cases.
True Negative (TN): Correctly predicted negative cases.
False Positive (FP): Incorrectly predicted positive cases (Type I Error).
False Negative (FN): Incorrectly predicted negative cases (Type II Error).

1) Accuracy: the term accuracy calculates how many total predictions were correct by using

$$accuracy = \frac{TP+TN}{TP+FN+FP+FN}$$

2) Precision: calculate how many of the predicted positives were correct Using

$$Precision = \frac{TP}{TP+FP}$$

Inception-ResNet-v2 [14]: is a deep convolutional neural network that merges the Inception architecture with ResNet-style residual connections to improve both learning speed and accuracy. The model consists of a stem module for initial feature extraction, followed by repeated blocks of Inception-ResNet-A, -B, and -C modules, each designed to process features at different spatial resolutions. These modules use parallel convolutional paths (like in Inception) and add the output back to the input via shortcut connections (like in ResNet). Between these blocks, reduction modules down sample the feature maps. The network ends with global average pooling and a fully connected layer for classification. This design allows the model to be very deep (over 50 layers) while maintaining good gradient flow, enabling high performance on image classification tasks like ImageNet.

Multi-tasks learning: Deep model is well suited for multi-task learning since the features learned from a task may be useful for other tasks. Existing multi-task deep models [24] are not suitable to solve our problems because they assume similar learning difficulties and convergence rates across all tasks.

3. Related work:

In recent years, researchers have made considerable progress in developing facial asymmetry detection. There are some works related to facial asymmetry and facial paralysis.

The approach based on the limited-orientation modified circular Gabor filters (LO-MCGFs) was proposed by Ngo et al. [15] for an objective and quantitative analysis of facial palsy. The LO-MCGFs use uniform passbands to re move noises and enhance the desired spatial frequencies, and use bounded filter support to specify the region of inter est. These virtues make the approach effective for extracting facial asymmetrical features. The facial expression dataset composed of 75 patients and 10 participants without facial palsy, made by the Osaka Police Hospital, was used in their experiments. Each expression is composed of 60 still images per subject and the intensity in each image is scored into 3 levels, strong, median and weak. As it is a proprietary dataset, it is not known whether the images are from continuous expression variation, and how the intensity level is assigned.

Gary Storey [16] proposed Face Symmetry Analysis Using Unified Multi-task CNN for Medical Applications. The authors propose a novel unified multi-task Convolutional Neural Network (CNN) framework that simultaneously performs object proposal, face detection, and face symmetry analysis. This integrated approach is particularly beneficial for medical applications, such as diagnosing and monitoring conditions like Bell's palsy, where facial symmetry plays a crucial role. The leverage of the CK+ database [17] for training and testing of our proposed methods. The database consists of 593 sequences containing 10,201 individual frames from 123 subjects. They received 98.3% accuracy on ck+. In testing they use 23 facial palsies set the set contain 25 images, the result was an issue the model classified 17 images as symmetrical out of 25.

Wang et al. [25] proposed an algorithm using ASM considering both static and dynamic facial asymmetries for recognizing facial movement patterns from facially paralyzed patients. But the intensity of FNP has not been quantitatively evaluated. In this study, video data from 62 patients with varying degrees of facial nerve paralysis (graded by the House–Brackmann scale) were used. Patients performed facial movements like smiling and eye closure, from which both static (mirror symmetry) and dynamic (motion speed) features were extracted. Using both types of features, the method achieved a classification accuracy of 97.56%, outperforming static-only analysis (93.52%). This approach offers a reliable tool for clinical evaluation of facial paralysis.

4. Method:

My project is about finding deep learning methods that fit under these three conditions working on images data, they use CNN as their based model and working on 2D dataset. I will present the 4 methods that I chose to make a comparison between them.

1. Four-layer ConvNet to facial emotion recognition with minimal epochs and the significance of data diversity [23]: This method is monopolized by Tanoy Debnath1, Md. Mahfuz Reza1 etc. The purpose of the research is to recognize facial emotions using face images, the method based on CNN and the images are 2D dataset. Model overview: they aim to create four layers of convolution together with two layers that are fully connected as shown in fig.1. map.

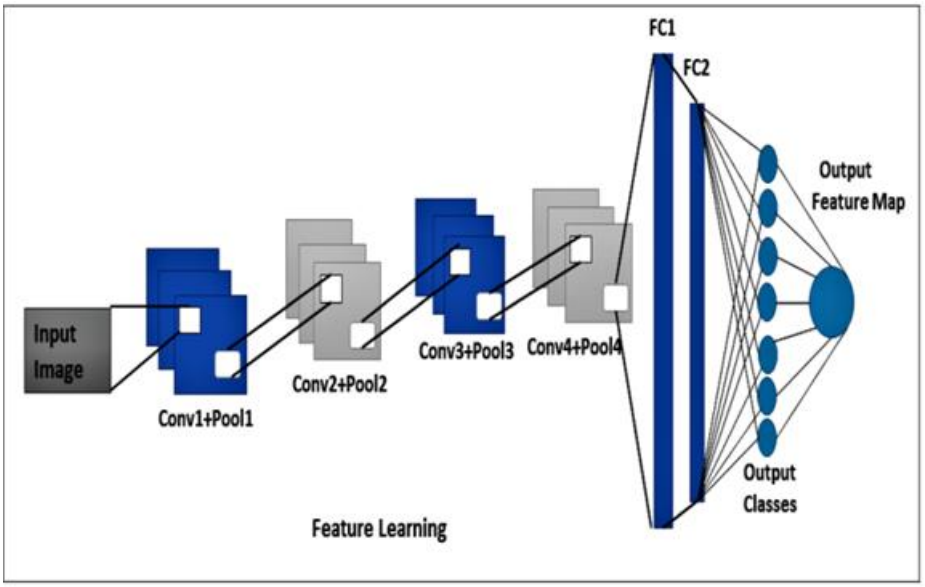


Fig.1. The graphical representation of the proposed CNN model for facial expression recognition

In addition, the convolution layer has weights that need to be taught, whereas the pooling layers are used as a fixed function to convert the activation. The Rectified Linear Unit (ReLU) is used to provide non-linearity to the entire network while having no effect on the convolutional layer's receptive fields. Figure1. The graphical representation of the proposed CNN model for facial expression recognition. Pipeline for proposed CNN. The network with a layer for processing the input. Here are four convolution and additional pooling layers and two fully connected layers that are completely associated at the end. A ReLU layer, batch normalization, and a dropout layer are used for any convolution and a fully connected layer of all the four network structures. The additional dense layer is used at the end of the four convolution layers which are associated with the two fully connected layers.

2. Visual Facial Paralysis Detection using InceptionResNetV2[14]: This method is monopolized by Nurul Natasha Razlan, Raihah Aminuddin etc. The purpose of this research is to detect facial paralysis using face images,

they use CNN and InaeptionResNetV2. The architecture comprises distinct layers, including inception blocks, residual connections, and global average pooling, aiming to enrich its understanding and classification capabilities. It begins with an input image of dimensions (224x224x3) as shown in fig3.

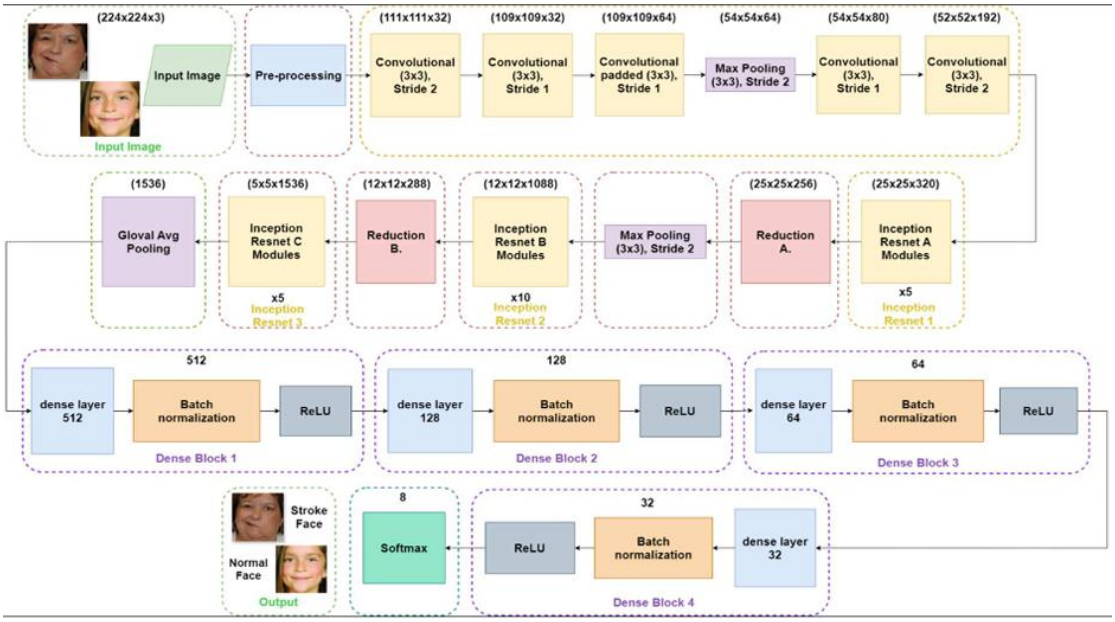


Fig. 3. Facial Paralysis Detection using InceptionResNetV2

3. Facial expression recognition via deep learning [22]: This method is monopolized by Abir Fathallah, Lotfi Abdi and Ali Douik. This method made to recognize facial expression. They aim to ameliorate the accuracy of facial expression classification by using a new CNN architecture. Model overview: they build four Convolutional layers with 3 pooling layers and two full connected layers as shown in fig4.

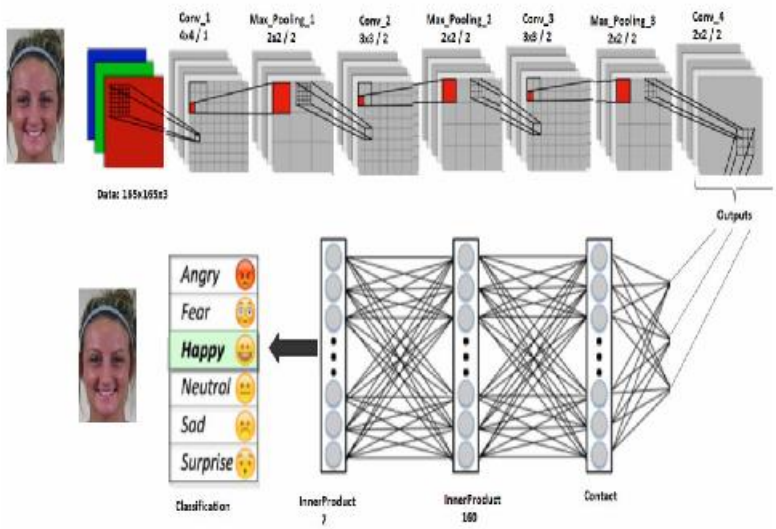


Fig. 4. Architecture of the convolutional neural network.

4. Facial Palsy Detection Through Changes in Facial Muscle Functionality Using CNN Algorithm [21]: This method is monopolized by Chindy Aulia Sari, Fitra Abdurrachman Bachtiar and Lailil Muflikhah. The purpose of this research is to detect facial palsy using CNN algorithm. Model overview: Process feature learning on the proposed CNN architecture will use 2 convolution layers with The ReLU activation function on each layer, the type of pooling layer that will be used is MaxPooling. After the feature learning process, the data will be converted into vector using flatten layer. Furthermore, the data will go through the fully connected layer for the classification process. As shown in fig5.

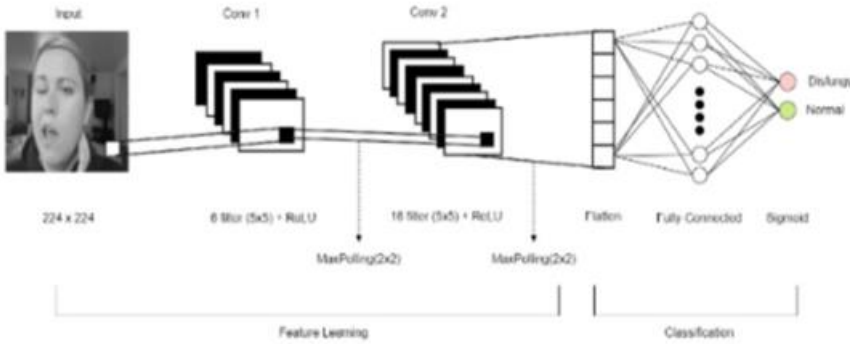


Fig. 5. Illustration of CNN architecture implementation

To detect the best face region that identifies facial asymmetry better I will use prebuild model MTCNN that is available at facenet package. To achieve the goal.

1) Detect face from every image.
2) All the cropped images are saved at drive as a new data set.
3) For every image in cropped face dataset split the image lower part and upper part.
4) Save both parts as a new dataset in drive.

5. Data:

A dataset consisting of 2825 face images with varying amounts of facial palsy has been collected. The dataset includes palsy-affected face images collected from Kaggle source [20]. In addition, already classified data has been collected before. In this experiment I built 2 data sets, the first include 2 classes and the second 6 classes.

1. 2 classes dataset: the purpose of building this dataset is to reach the best for the Visual Facial Paralysis Detection using InceptionResNetV2[14] model. The dataset includes two classes symmetrical face and asymmetrical face. 1005 images of symmetrical faces and 1820 images of asymmetrical faces.
2. 6 classes dataset: the other models are going to use this dataset. The dataset includes 6 classes as shown in fig6.

Grade	Class	Size
I	Normal	501
II	Near Normal	80
III	Mild Dysfunction	542
IV	Moderate Dysfunction	468
V	Sever Flaccid	720
VI	Complete Flaccid	514

Fig.6. table to describe the 6 classes dataset.

6. Experiment

the experiment done using Google Colab's tools while the datasets are in drive.

1. Data Preprocessing The preprocessing stage was carried out after secondary data was rearranged into normal and dysfunctional classes. In this stage, the dataset is split into training and testing data. The training data is utilized for modeling purposes, while the testing data is reserved for testing and analysis.
2. The dataset is divided with an 80% proportion for training data and a 20% proportion for testing data. This split ratio has been commonly used as a benchmark for training and evaluating machine learning models in the field of image recognition.
   After the data has been divided into training and testing sets, the image data undergoes essential transformations before being utilized in the training and testing process. Due to the conflict on the input image's size for every method there will be four transforms one for each method.

Methods	Tanoy Debnath [23]	Nurul Natasha Razlan [14]	Abir Fathallah [22]	Chindy Aulia Sari [21]
Input size	48x48x3	224x224x3	165x165x3	224x224x1

3. Train and test the models:
   Due to the limited usage of Google Colab's tools and GPU, the epochs number small, this will put some models like Nurul Natasha [2] at a disadvantage position. The process will be: The process begins by initializing a convolutional neural network model and moving it to the appropriate device (CPU or GPU). The loss function used is CrossEntropyLoss, suitable for multi-class classification, and the optimizer is Adam with a different learning rate for each model. The model is trained for certain number of epochs using training data. In each epoch, the model is set to training mode, and for each batch of images and labels, the gradients are reset, predictions are made, loss is calculated, and weights are updated through backpropagation. Training accuracy and average loss per epoch are printed to monitor learning progress. After training, the model

is switched to evaluation mode and tested on unseen test data without computing gradients. The test accuracy is calculated by comparing the model's predictions to the true labels. This helps assess how well the model generalizes to new data. Except for the Chindy Aulia Sari [21] model all the other models train and test like that.
Chindy Aulia Sari [21] uses the k-fold training method, while k=5 I split the data into 5 different parts every 10 epochs. I chose 4 parts to be the train, and the last part will be the test part. In this case the model trains on every image 4 times and tests the image one time.

7. Results and discussion:

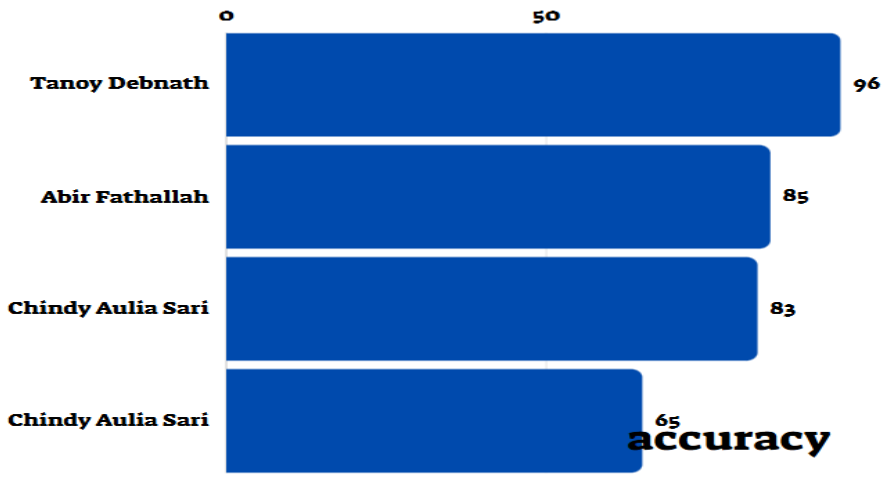
Based on the experimental results presented in this study, Tanoy Debnath [23] Four-layer ConvNet to facial emotion recognition with minimal epochs and the significance of data diversity method received best performance here is some key point why this happened.

1) Nurul Natasha Razlan's [14] model considers a complex model with many layers even after increasing the epochs number from 20 to 35 it still got 65% accuracy.

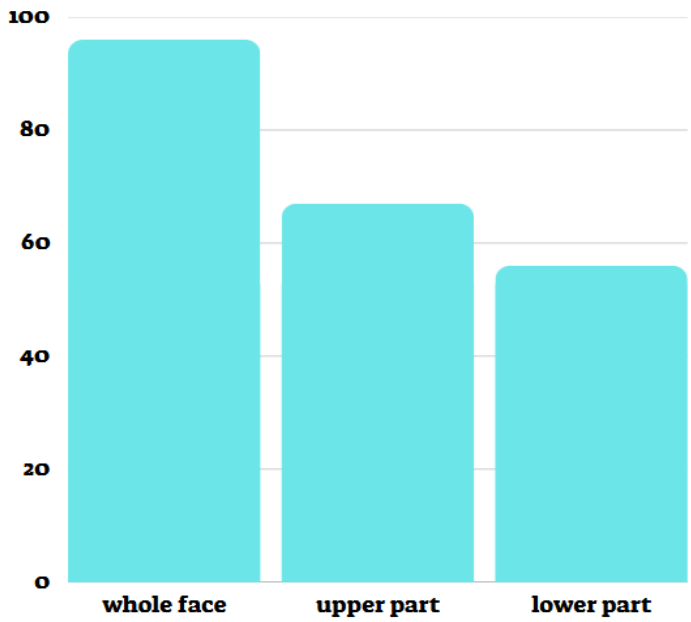
2) Chindy Aulia Sari's [21] model got a better performance than Nurul Natasha Razlan [14] with 83% accuracy. But the big difference here is that the model is very simple but because of the k-fold method it got better accuracy. But Tanoy Debnath [23] still got better accuracy because of the 2-classes dataset. The dataset contains the asymmetrical face (unhealthy face) and symmetrical face (healthy face) will there is 6 degrees of symmetry so it's hard to let the model choose about asymmetrical healthy face.

3) Abir Fathallah's [22] model is good but much more complex than Tanoy Debnath [23] even after 45 epochs with rl = 0.01 it got 85% accuracy.

For the Tanoy Debnath [23] model it receives 96% accuracy which is pretty good for simple model like that. I decreased the lr from 0.01 to 0.001 and added weight decay to prevent overfitting. Also, we have data augmentation for every model that will help to prevent overfitting too by making changes on the image like rotation and flip.



For the best part that identifies facial asymmetry better the result is obvious the whole face dataset received 96.46% accuracy while the lower face dataset received 67% and upper face dataset received 56%.



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